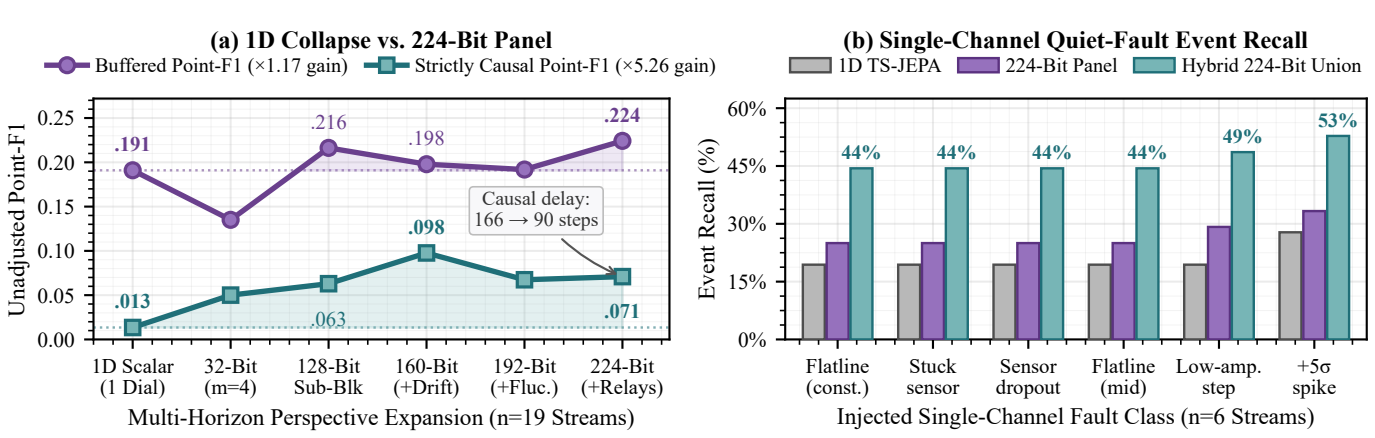

(a) 1D Collapse vs. 224-Bit Panel
Buffered Point-F1 (×1.17 gain)
Strictly Causal Point-F1 (×5.26 gain)
Unadjusted Point-F1
0.25
0.20
0.15
0.10
0.05
0.00
.191
.216
.198
.224
.013
.063
.098
.071
Causal delay:
166 → 90 steps
1D Scalar (1 Dial)
32-Bit (m=4)
128-Bit Sub-Blk
160-Bit (+Drift)
192-Bit (+Fluc.)
224-Bit (+Relays)
Multi-Horizon Perspective Expansion (n=19 Streams)
(b) Single-Channel Quiet-Fault Event Recall
1D TS-JEPA
224-Bit Panel
Hybrid 224-Bit Union
Event Recall (%)
60%
45%
30%
15%
0%
44%
44%
44%
44%
49%
53%
Flatline (const.)
Stuck sensor
Sensor dropout
Flatline (mid)
Low-amp. step
+5σ spike
Injected Single-Channel Fault Class (n=6 Streams)